

Lecture 2 handout: Beta-binomial conjugacy

Analytic derivation moved out of the lecture slides

Statistics 367-1-4361

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1 Why this handout exists

In the lecture, the main point of *conjugate priors* is conceptual:

- a Beta prior is a natural prior for an unknown probability,
- a binomial likelihood updates that prior analytically, and
- the posterior stays in the Beta family.

The algebra needed to prove that result is useful, but it is not the best use of slide time. This handout contains the derivation carefully so that the lecture slides can stay focused on updating, interpretation, and intuition.

2 Model and notation

We observe z heads in N coin tosses, and we want to learn about the unknown probability of heads θ .

We use the model

$$\theta \sim \text{Beta}(\alpha, \beta)$$

and

$$z \mid \theta \sim \text{Binomial}(N, \theta).$$

Here:

- $\theta \in [0, 1]$ is the probability of heads,
- $\alpha, \beta > 0$ are prior shape parameters,
- z is the number of heads,
- $N - z$ is the number of tails.

3 Step 1: Write Bayes' rule

Bayes' rule says

$$p(\theta \mid z, N) = \frac{p(z \mid \theta, N) p(\theta)}{p(z \mid N)}.$$

The denominator

$$p(z \mid N) = \int_0^1 p(z \mid \theta, N) p(\theta) d\theta$$

is the **evidence** (also called the marginal likelihood). It is the normalizing constant that makes the posterior integrate to 1.

For the derivation, it is easiest to start with the proportional form

$$p(\theta \mid z, N) \propto p(z \mid \theta, N) p(\theta).$$

4 Step 2: Write the prior and likelihood explicitly

The Beta prior density is

$$p(\theta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}, \quad 0 \leq \theta \leq 1,$$

where

$$B(\alpha, \beta) = \int_0^1 \theta^{\alpha-1} (1 - \theta)^{\beta-1} d\theta$$

is the Beta function.

The binomial likelihood is

$$p(z \mid \theta, N) = \binom{N}{z} \theta^z (1 - \theta)^{N-z}.$$

5 Step 3: Multiply prior and likelihood

Now multiply them:

$$p(z \mid \theta, N) p(\theta) = \binom{N}{z} \theta^z (1 - \theta)^{N-z} \cdot \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}.$$

Group the constants together and combine powers of θ and $(1 - \theta)$:

$$p(z \mid \theta, N) p(\theta) = \frac{\binom{N}{z}}{B(\alpha, \beta)} \theta^{z+\alpha-1} (1 - \theta)^{N-z+\beta-1}.$$

Since the factor

$$\frac{\binom{N}{z}}{B(\alpha, \beta)}$$

does not depend on θ , we can absorb it into the proportionality constant. Therefore

$$p(\theta \mid z, N) \propto \theta^{\alpha+z-1} (1-\theta)^{\beta+N-z-1}.$$

6 Step 4: Recognize the kernel of a Beta distribution

A $\text{Beta}(a, b)$ density has the form

$$p(\theta) \propto \theta^{a-1} (1-\theta)^{b-1}.$$

Comparing this with

$$p(\theta \mid z, N) \propto \theta^{\alpha+z-1} (1-\theta)^{\beta+N-z-1},$$

we immediately identify

$$a = \alpha + z, \quad b = \beta + N - z.$$

So the posterior must be

$$\theta \mid z, N \sim \text{Beta}(\alpha + z, \beta + N - z).$$

This is the conjugacy result.

7 Step 5: Show the normalizing constant explicitly

To make the posterior a proper density, we divide by the Beta-function normalizing constant for the updated parameters:

$$p(\theta \mid z, N) = \frac{1}{B(\alpha + z, \beta + N - z)} \theta^{\alpha+z-1} (1 - \theta)^{\beta+N-z-1}.$$

Equivalently, the evidence is

$$p(z \mid N) = \int_0^1 p(z \mid \theta, N) p(\theta) d\theta = \frac{\binom{N}{z}}{B(\alpha, \beta)} \int_0^1 \theta^{\alpha+z-1} (1 - \theta)^{\beta+N-z-1} d\theta.$$

The integral is exactly a Beta function:

$$\int_0^1 \theta^{\alpha+z-1} (1 - \theta)^{\beta+N-z-1} d\theta = B(\alpha + z, \beta + N - z).$$

So the evidence is

$$p(z \mid N) = \frac{\binom{N}{z} B(\alpha + z, \beta + N - z)}{B(\alpha, \beta)}.$$

You do not need this expression to identify the posterior family, but it is the missing normalization step in the full derivation.

8 Interpretation of the update

The posterior update is

$$\text{Beta}(\alpha, \beta) \xrightarrow{z \text{ heads in } N \text{ tosses}} \text{Beta}(\alpha + z, \beta + N - z).$$

So the data update the parameters directly:

- add the number of heads to α ,
- add the number of tails to β .

This is why conjugacy is so attractive pedagogically: the update rule is exact, compact, and easy to interpret.

9 Posterior summaries

Once we know the posterior is $\text{Beta}(\alpha + z, \beta + N - z)$, we can read off useful summaries.

9.1 Posterior mean

$$\mathbb{E}[\theta \mid z, N] = \frac{\alpha + z}{\alpha + \beta + N}.$$

9.2 Posterior mode

When $\alpha + z > 1$ and $\beta + N - z > 1$,

$$\text{mode}(\theta \mid z, N) = \frac{\alpha + z - 1}{\alpha + \beta + N - 2}.$$

9.3 Posterior variance

$$\text{Var}(\theta \mid z, N) = \frac{(\alpha + z)(\beta + N - z)}{(\alpha + \beta + N)^2(\alpha + \beta + N + 1)}.$$

These formulas make the next lecture section on posterior updating much easier to interpret.

10 Sequential updating

Suppose we observe the tosses one at a time. If the current posterior is

$$\theta \mid y_{1:n} \sim \text{Beta}(\alpha_n, \beta_n),$$

then the next toss updates as follows:

- if $y_{n+1} = 1$ (a head), then

$$\alpha_{n+1} = \alpha_n + 1, \quad \beta_{n+1} = \beta_n;$$

- if $y_{n+1} = 0$ (a tail), then

$$\alpha_{n+1} = \alpha_n, \quad \beta_{n+1} = \beta_n + 1.$$

This makes the repeated posterior-updating plots in the lecture completely natural.

11 What to remember

The main lesson is not the integral itself. The main lesson is the structure:

1. start with a prior family,
2. multiply by the likelihood,
3. recognize the kernel of a known distribution,
4. normalize.

For the Beta prior and binomial likelihood, this process produces another Beta distribution:

$$\theta \mid z, N \sim \text{Beta}(\alpha + z, \beta + N - z).$$

That is exactly why the Beta prior is called **conjugate** to the binomial likelihood.